MACHINE LEARNING ALGORITHMS

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Session 1 –Topics

- What is Machine Learning?
- Machine Learning Applications
- Major Machine Learning Techniques
- Introduction to python Libraries for Machine learning
- Difference between AI vs ML vs DL
- Types of Machine Learning
- Data Pre-processing (Hands-on)

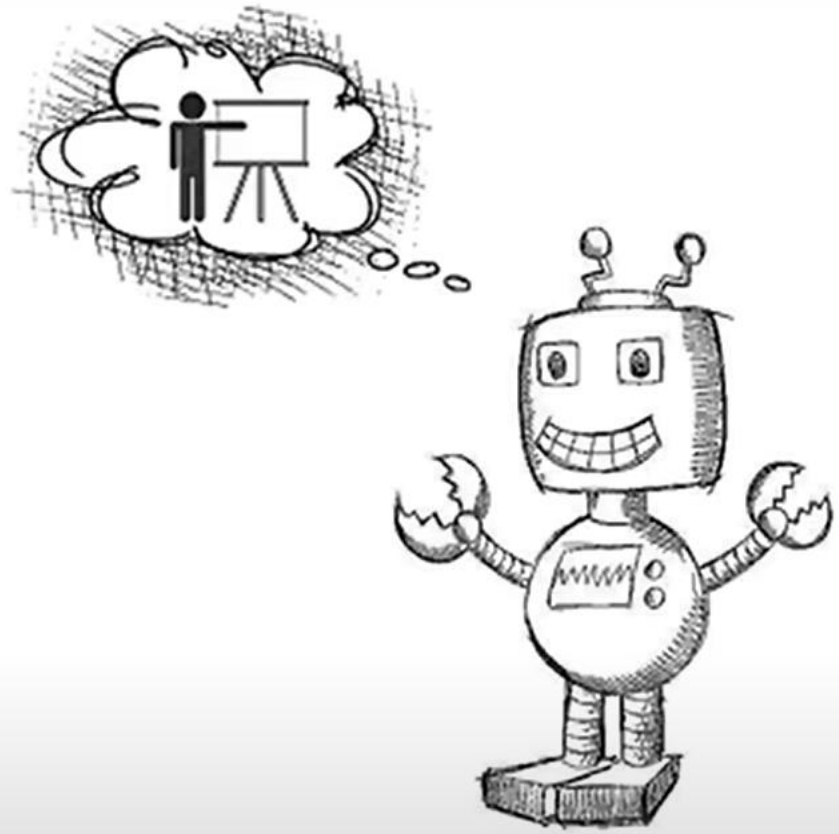
"Why is machine learning more popular now, even though it was first developed in the 1960s?"

Ans:

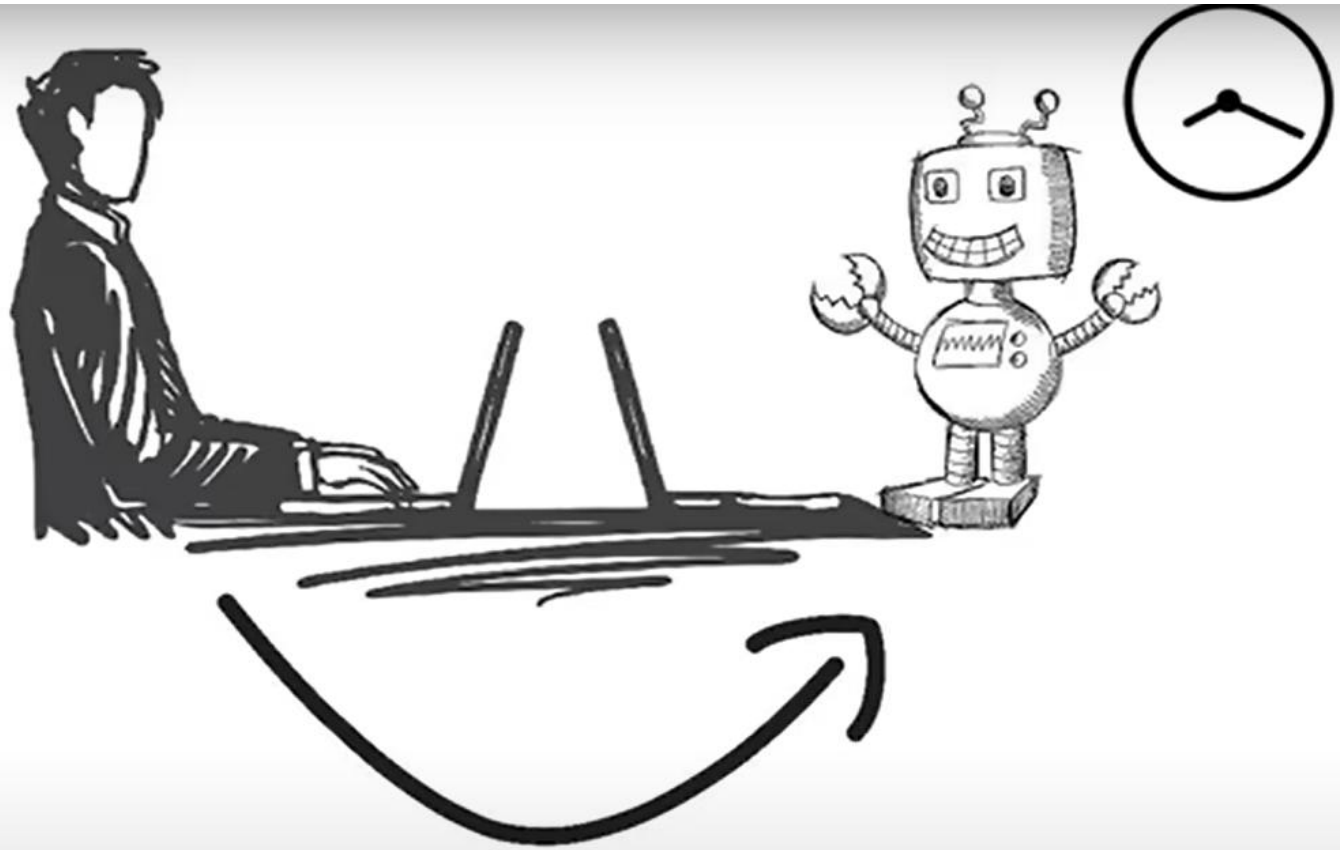
- The rise of **Big data** and Internet
- Improved computation power.
- New algorithms and techniques,to better handle and learn from data



HUMANS LEARN FROM
PAST EXPERIENCES



MACHINES FOLLOW INSTRUCTIONS
GIVEN BY HUMANS



WHAT IF HUMANS CAN TRAIN THE MACHINES...

TO LEARN FROM THE PAST DATA AND DO WHAT HUMANS CAN DO AND MUCH FASTER THAT'S CALLED MACHINE LEARNING

What is machine learning?

Machine learning is the subfield of computer science that gives "**computers the ability to learn without being explicitly programmed.**"

What is explicitly programmed?

In traditional programming, a developer writes out code that gives specific instructions that tell the computer exactly what to do in every situation.

For instance, if you were programming a calculator, you would explicitly code how to handle addition, subtraction, multiplication, etc.

Without being explicitly programmed

Let us discuss with a scenario

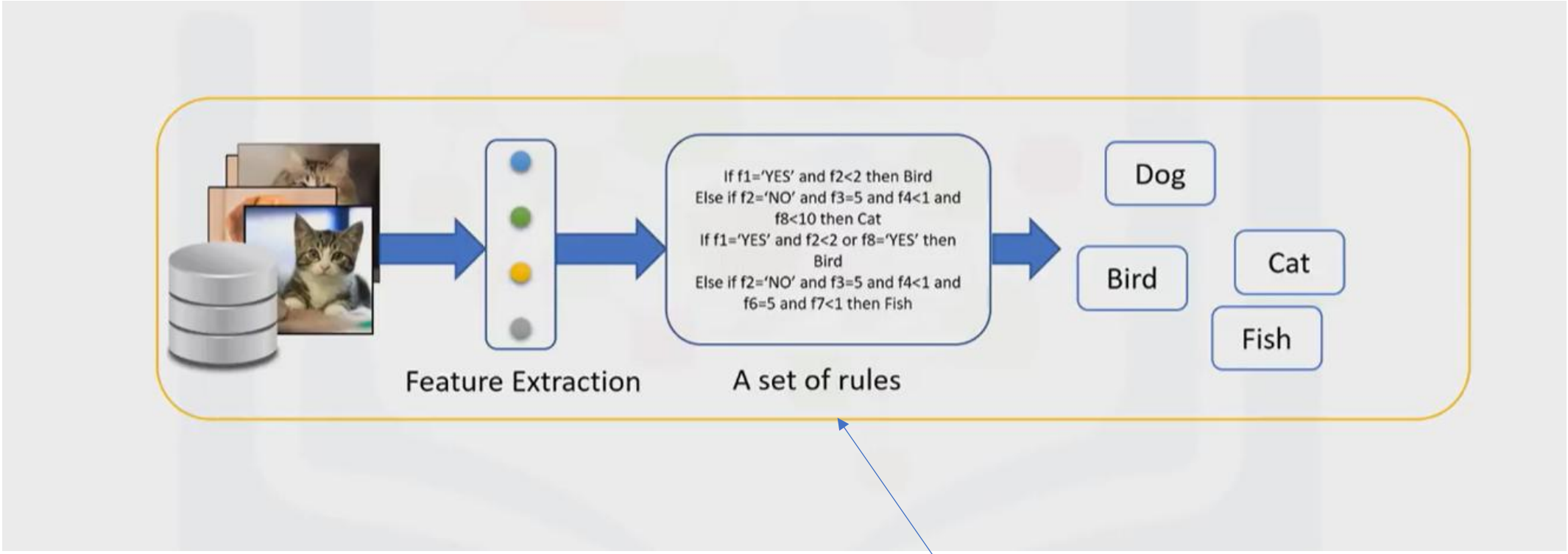
Assume that you have a **dataset of images** of **animals** such as **cats and dogs**, and you want to build a **software or an application** that can **recognize** and **differentiate them**.

……

The first thing that you have to do here is **interpret the images** as a **set of feature sets.**

For example,

- **Does the image show the animal's eyes?**
- **If so, what is their size?**
- **Does it have ears?**
- **What about a tail?**
- **How many legs?**
- **Does it have wings?**



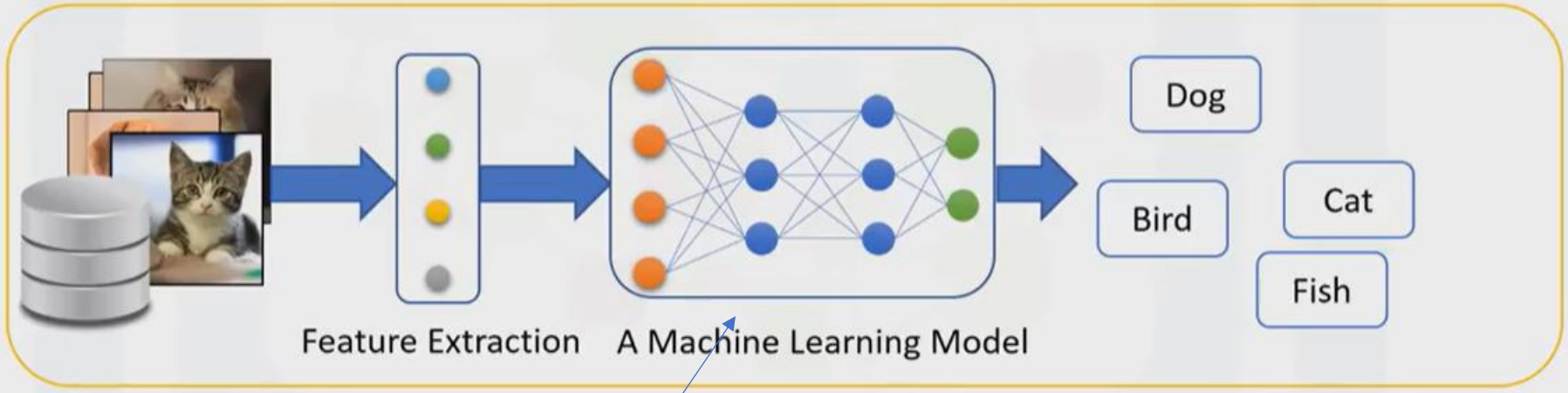
Prior to Machine learning traditionally, we had to write down **some rules or methods** in order to get our **application** to be **intelligent** and detect the animals.

But, it was a **failure.**

Why?

- It needed a lot of rules,
- Highly dependent on the current dataset,
- Not generalized enough to detect out-of-new sample datasets.

How machine learning works?

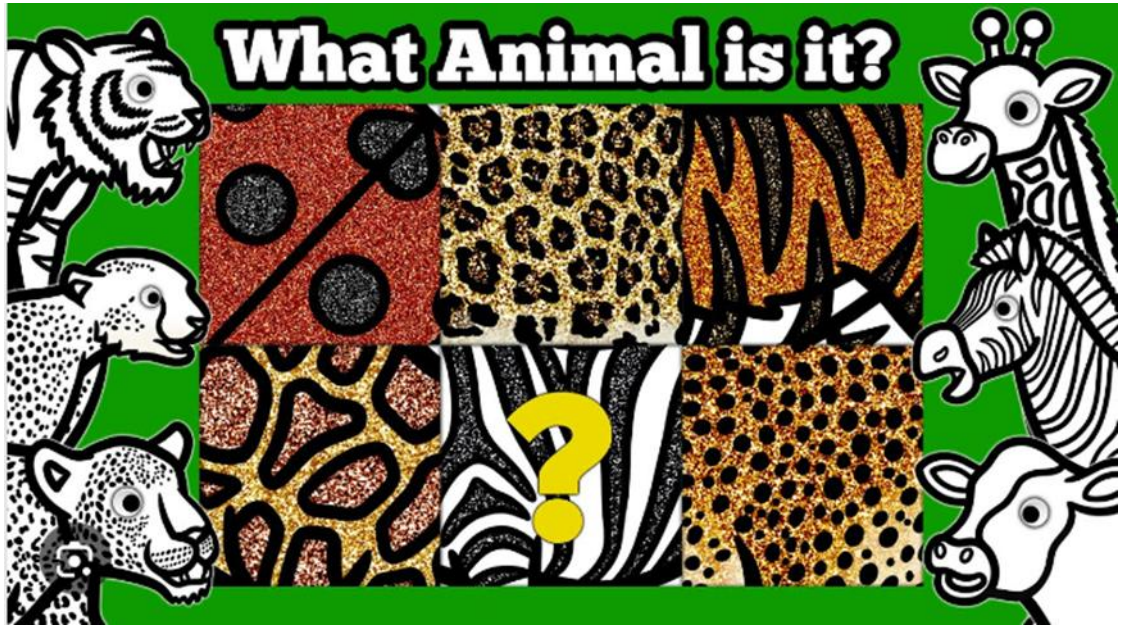


Machine learning allows us to **build a model** that looks at all the **feature sets**, and their corresponding type of animals, and learn the **pattern** of each animal.

- It is a model built by **machine learning algorithms**.
- It detects without **explicitly being programmed** to do so.

……

So, machine learning follows the same process that a **4-year-old child** uses to **learn, understand, and differentiate** animals.



So, **machine learning algorithms**, inspired by the **human learning process, iteratively learn from data**, and allow machines to find **hidden insights.**

Examples of Machine learning applications





How **Netflix and Amazon** **recommend videos, movies, and TV shows** to its users?

They use **Machine Learning** to produce **suggestions** that you **might enjoy!**

Examples of Machine learning applications



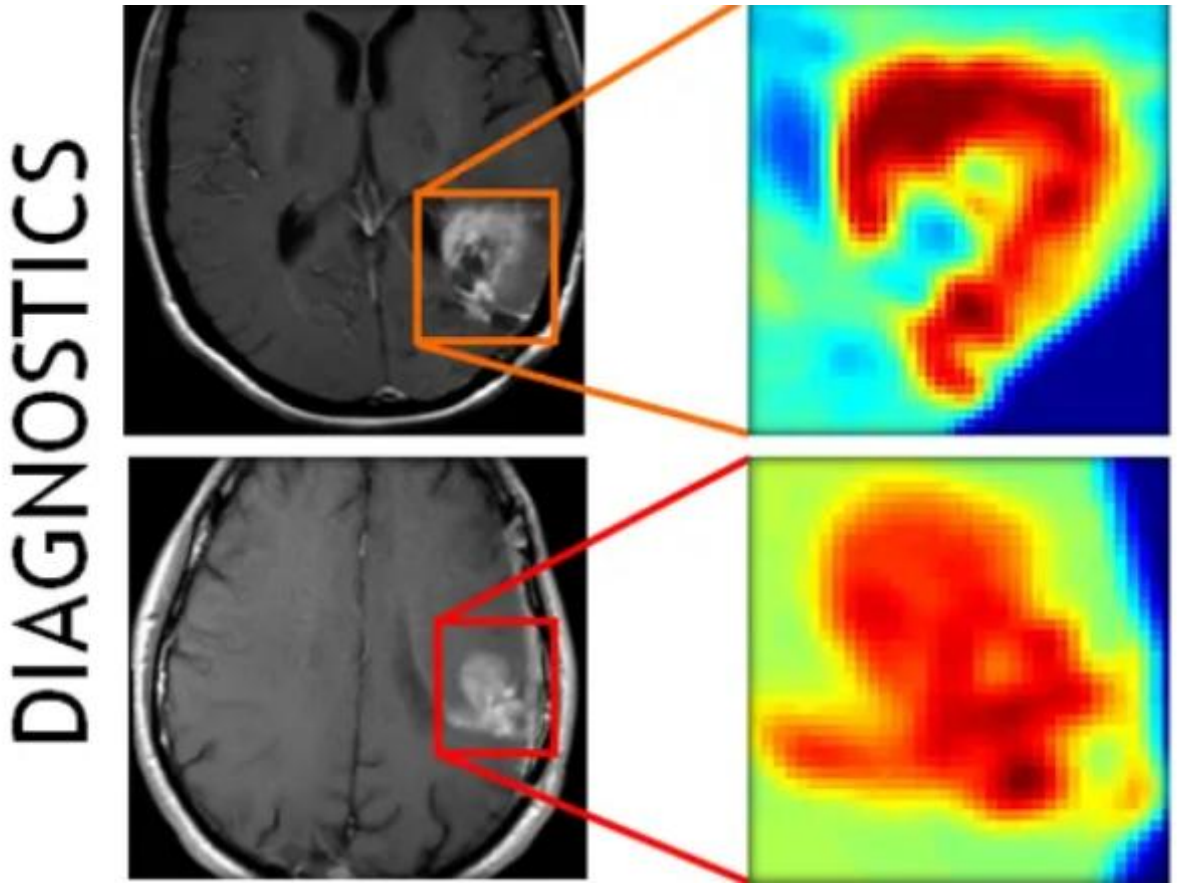
How do you think **banks make a decision** when **approving a loan application**?
They use **machine learning** to **predict the probability** of **default** for each applicant, and then **approve** or **refuse** the loan application based on that **probability.**



Telecommunication companies analyze their **customers' demographic data** to categorize them into **distinct groups** (segmentation) and to **forecast** which **customers** are likely to **cancel their service** (predict churn)

Examples of Machine learning applications

Healthcare Diagnostics



ML models analyze medical images (**X-rays** and **MRIs**) to assist in **diagnosing diseases**,1000 times faster than humans.

It also helps in

- ***Personalizing Treatment***
- ***Drug Discovery***
- ***Clinical Trial research***

Examples of Machine learning applications

Financial Trading



Autonomous vehicles

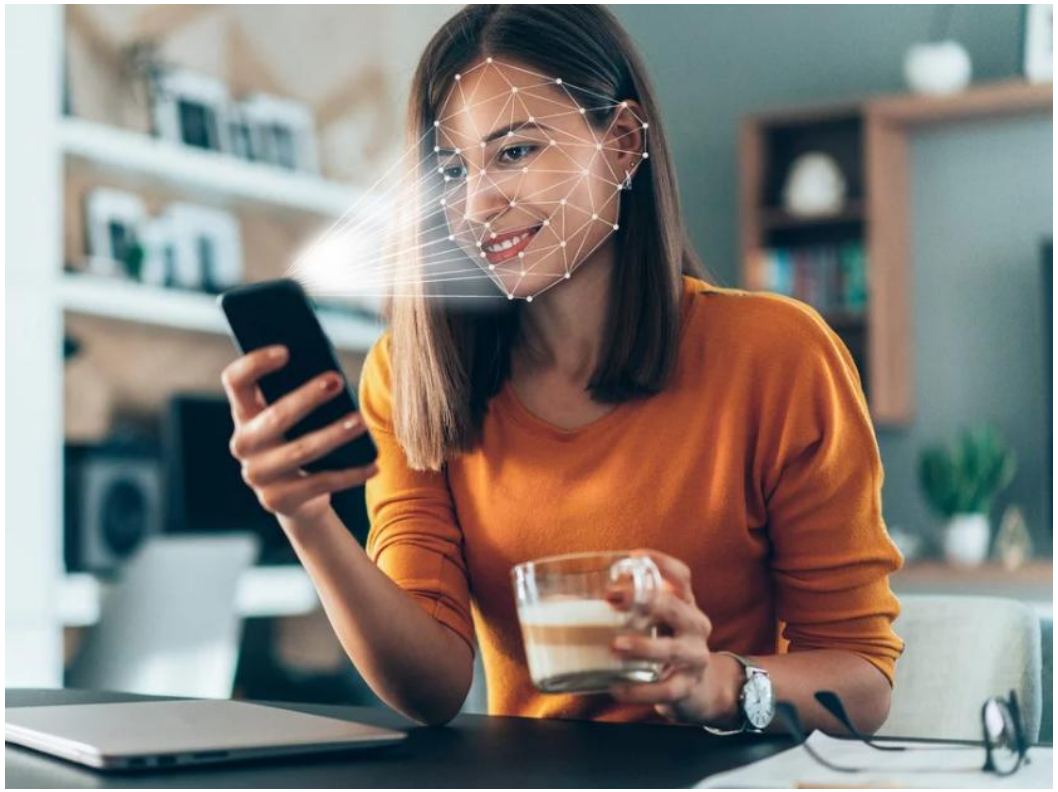


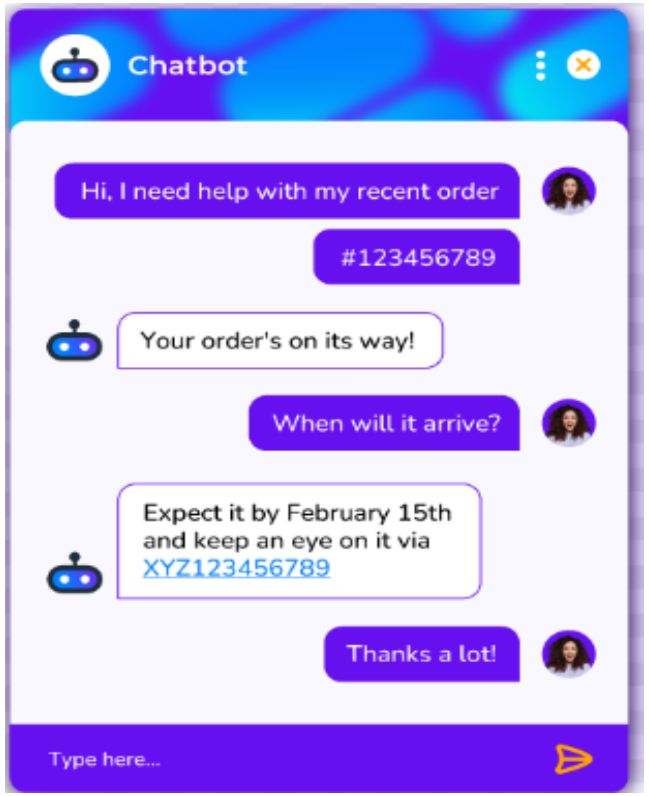
ML analyzes **market data** and **trends** to inform **trading strategies**, detect **Market shift , Risk assessment** and **predict stock prices**.

Self-driving cars use **ML** to **interpret sensor data**, **recognize objects**, **make real-time decisions**, and **navigate safely on the road.**

Examples of Machine learning applications

There are many other applications of machine learning that we see every day in our daily life, such as **chatbots**, **logging into our phones or even computer games using face recognition**.





Each of these use **different machine learning techniques and algorithms**.

Major machine learning techniques

- Regression/Estimation
  - Predicting continuous values

Predicting a **house's price** based on **features** like size, location, and number of bedrooms etc.

- Classification
  - Predicting the item class/category of a case

Churn prediction yes or no

- Clustering
  - Finding the structure of data; summarization

Grouping customers with similar buying habits

- Associations
  - Associating frequent co-occurring items/events

Market Basket Analysis

Major machine learning techniques

- Anomaly detection
  - Discovering abnormal and unusual cases
- Sequence mining
  - Predicting next events; click-stream (Markov Model, HMM)
- Dimension Reduction
  - Reducing the size of data (PCA)
- Recommendation systems
  - Recommending items

Credit card Fraud detection

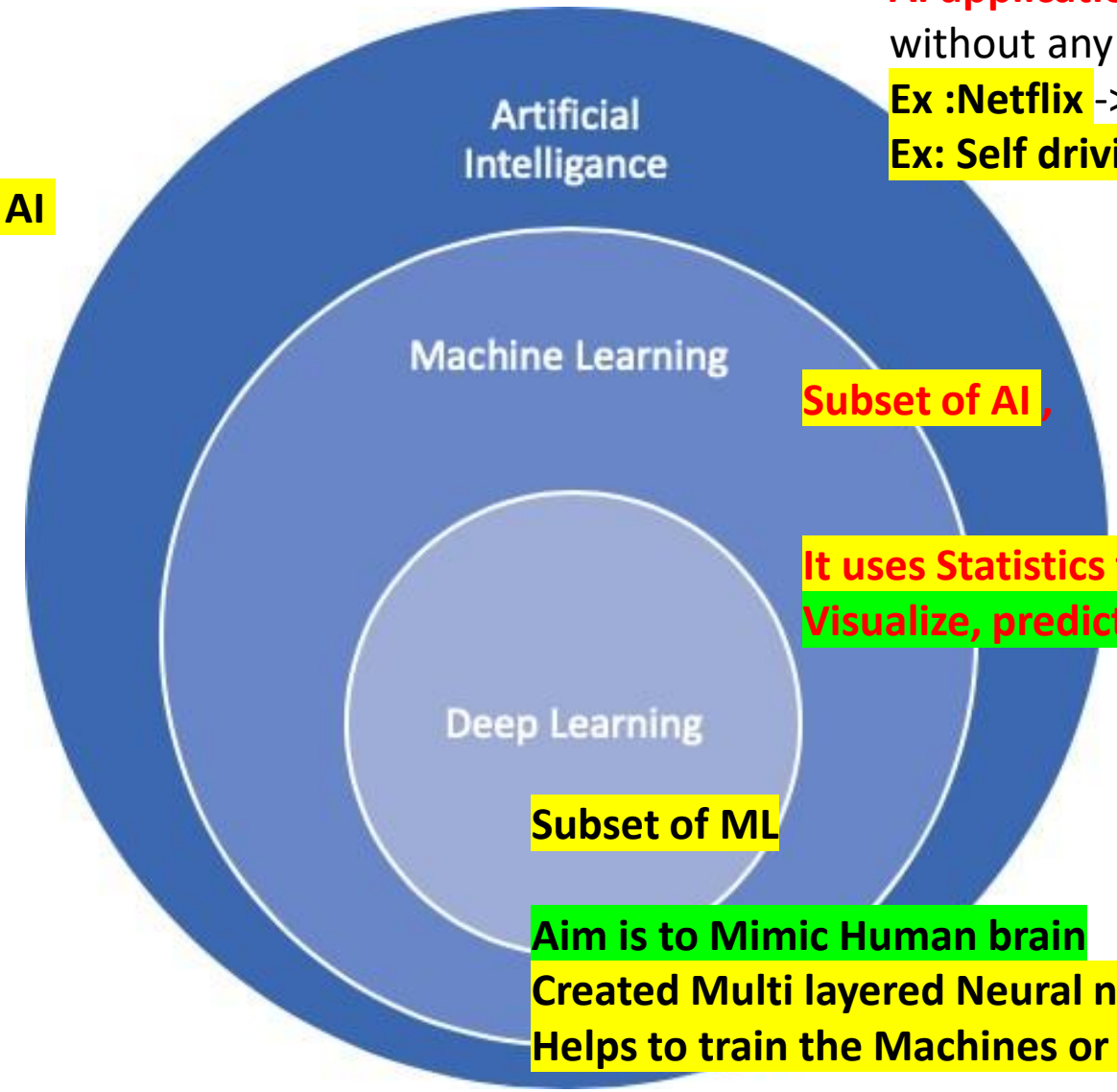
click-stream analysis on websites, it predicts the **next likely page** a user will visit based on their previous clicks.

Techniques like **Principal Component Analysis (PCA)** **used to reduce** the number of **features or variables** in a dataset while retaining as much **relevant information** as possible.

This associates people's **preferences** with others who have **similar tastes**, and **recommends new items** to them, such as **books or movies.**

Difference between AI & ML & DL

Whether you're a
ML Developer
DL Developer
CV Developer
AI Engineer
End of the day your creating **a AI application**



AI application is able to do it's own **task** without any **human intervention**.
Ex :Netflix -> **Movie Genre** -> **Recommendation**
Ex: Self driving cars

Subset of AI ,

It uses Statistics tools to analyze the data
Visualize, prediction, forecasting

Subset of ML

Aim is to Mimic Human brain
Created Multi layered Neural network
Helps to train the Machines or applications



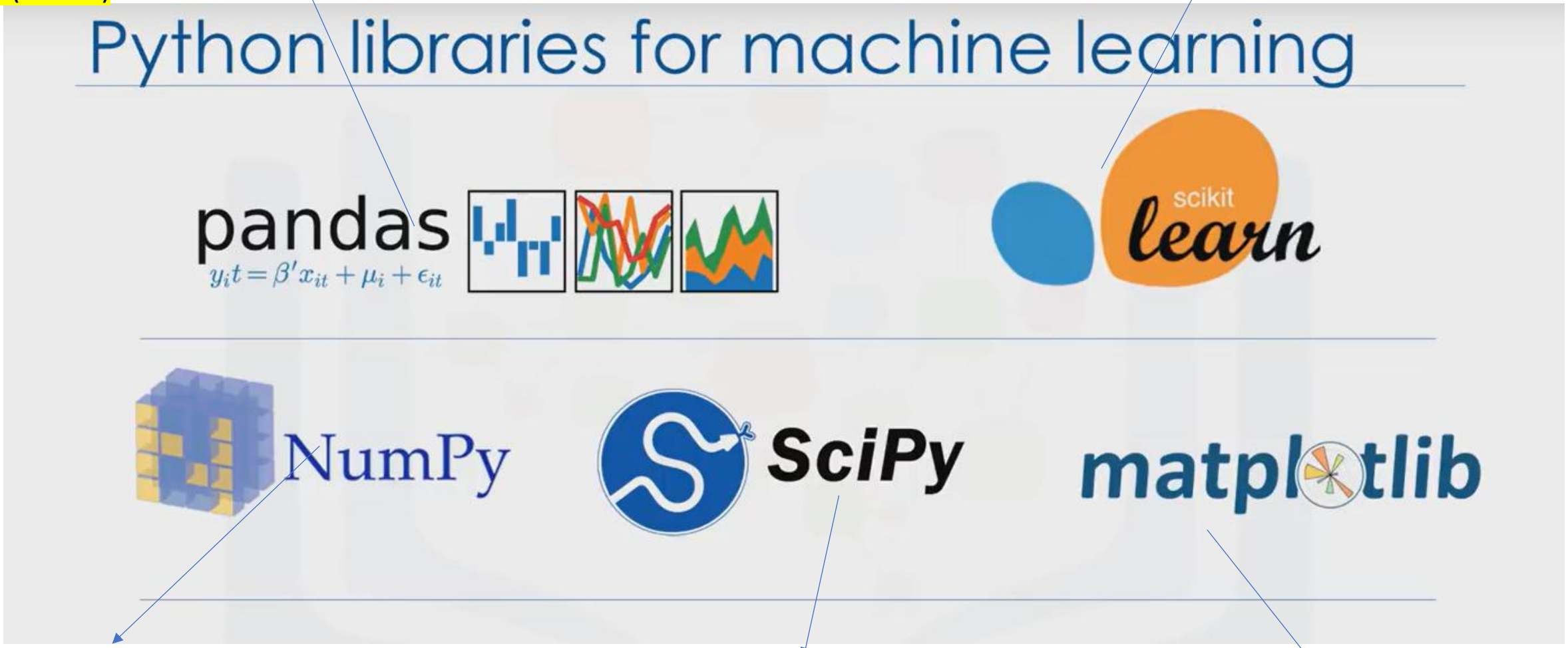
Python is a **popular and powerful general-purpose** programming language that recently emerged as the **preferred language** among **data scientists**.

You can write your **machine learning algorithm** using **python**, and it works very well.

However, there are a lot of **modules and libraries already implemented in python** that can make our work much easier.

For **data manipulation and analysis**, it provides powerful **data structures** like **Data Frames** for handling structured data(Tables).

simple and efficient tools for **data analysis**, including **classification, regression, clustering**, and more.



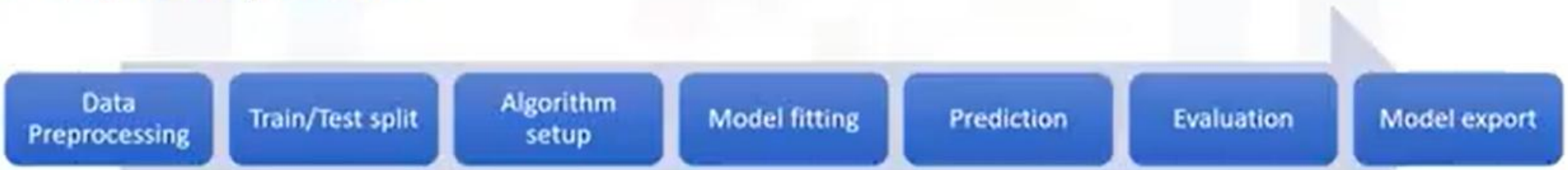
It is a **math library** to work with **n-dimensional arrays.**
It enables you to do computation efficiently and effectively.

It has **numerical algorithms** and **tools** for various domains like **signal processing**, **optimization, and statistics.**

Creating **interactive, animated visualizations, 2-D and 3-D plots.**

More about scikit-learn

- Free software machine learning library
- Classification, Regression and Clustering algorithms
- Works with NumPy and SciPy
- Great documentation
- Easy to implement



Most of the tasks that need to be done in a **machine learning pipeline** are implemented already in **scikit learn**

scikit-learn functions

Preprocess -Dataset

```
from sklearn import preprocessing
X = preprocessing.StandardScaler().fit(X).transform(X)
```

Split Data set into **Train and test**

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33)
```

```
from sklearn import svm
clf = svm.SVC(gamma=0.001, C=100.)
```

Set up the Algorithm

```
clf.fit(X_train, y_train)
```

Train the model with Training Data set

```
clf.predict(X_test)
```

Test the model with Test data set

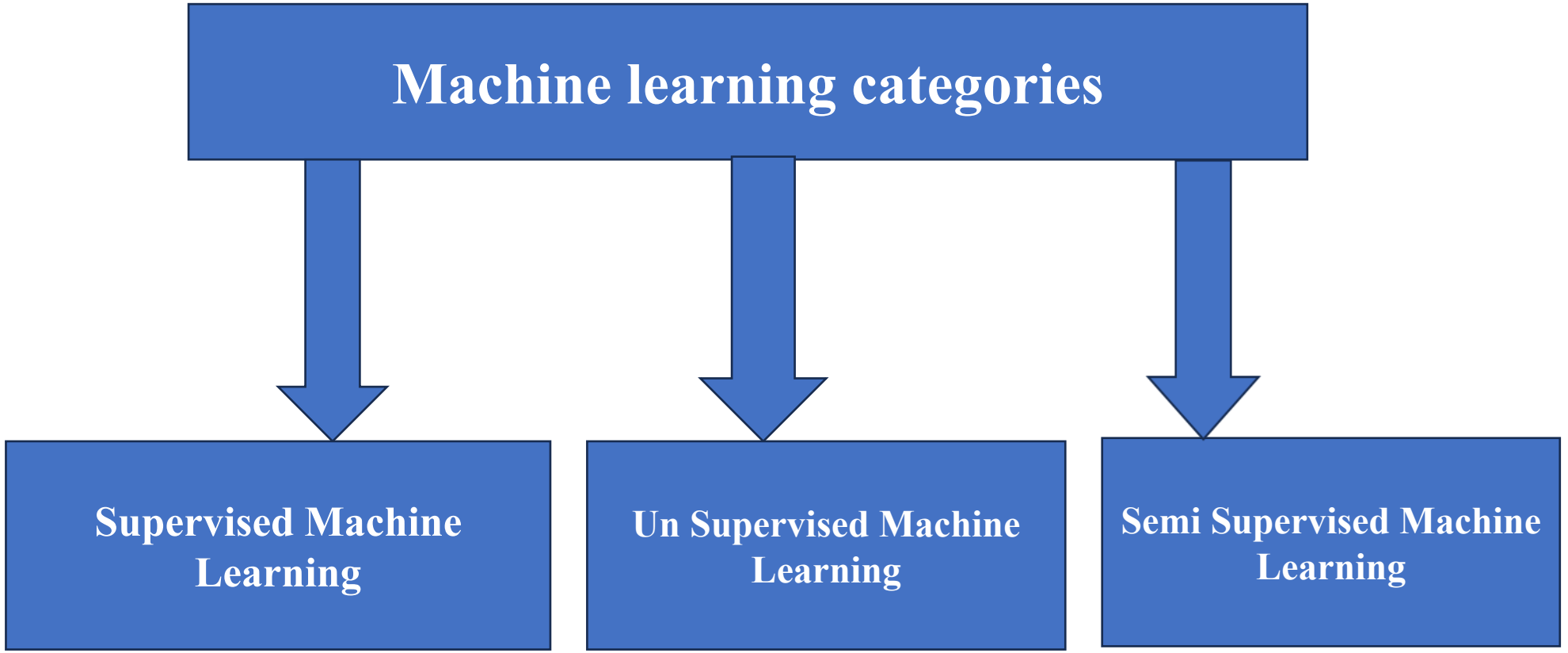
```
from sklearn.metrics import confusion_matrix
print(confusion_matrix(y_test, yhat, labels=[1,0]))
```

Evaluate the model accuracy and show the result

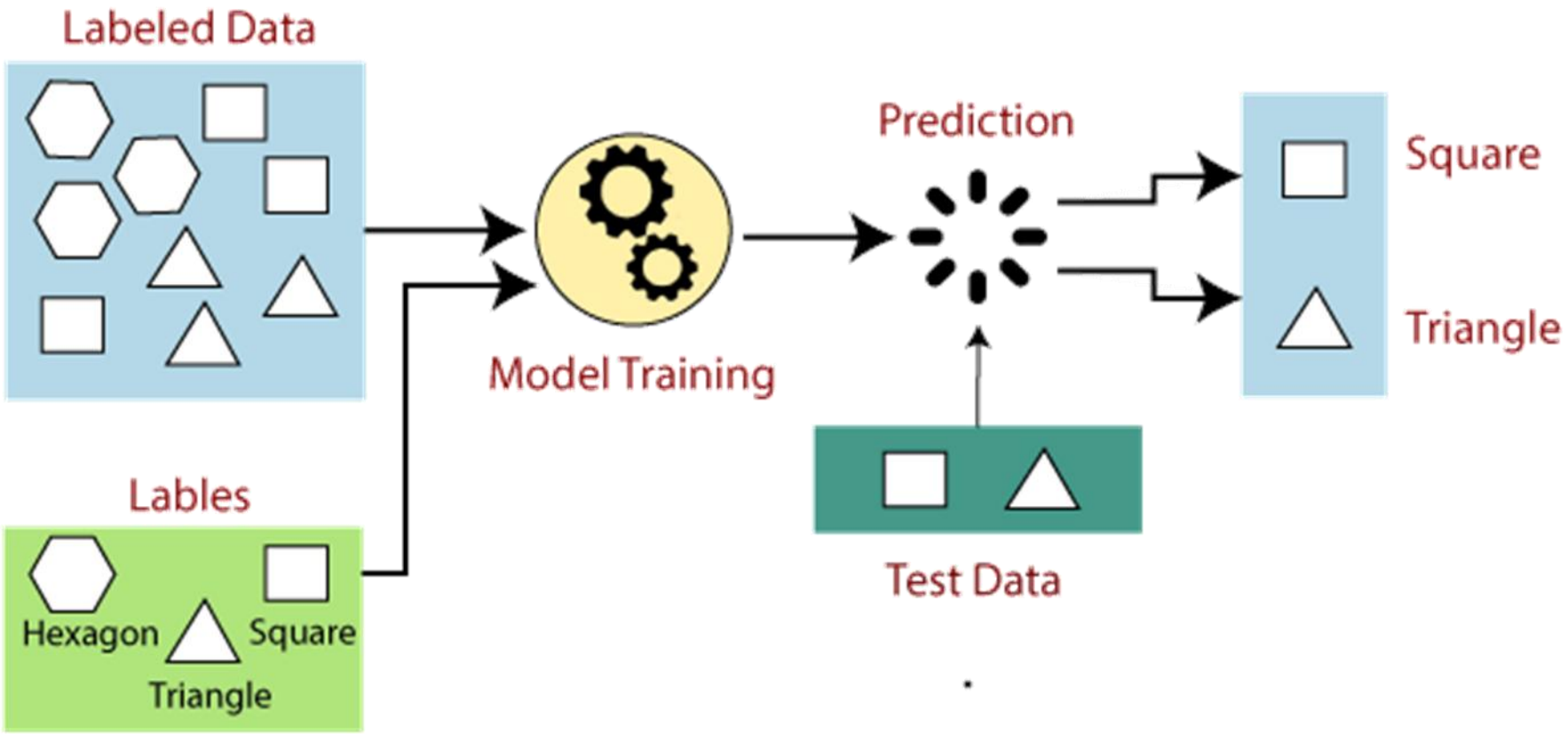
```
import pickle
s = pickle.dumps(clf)
```

Save the Model

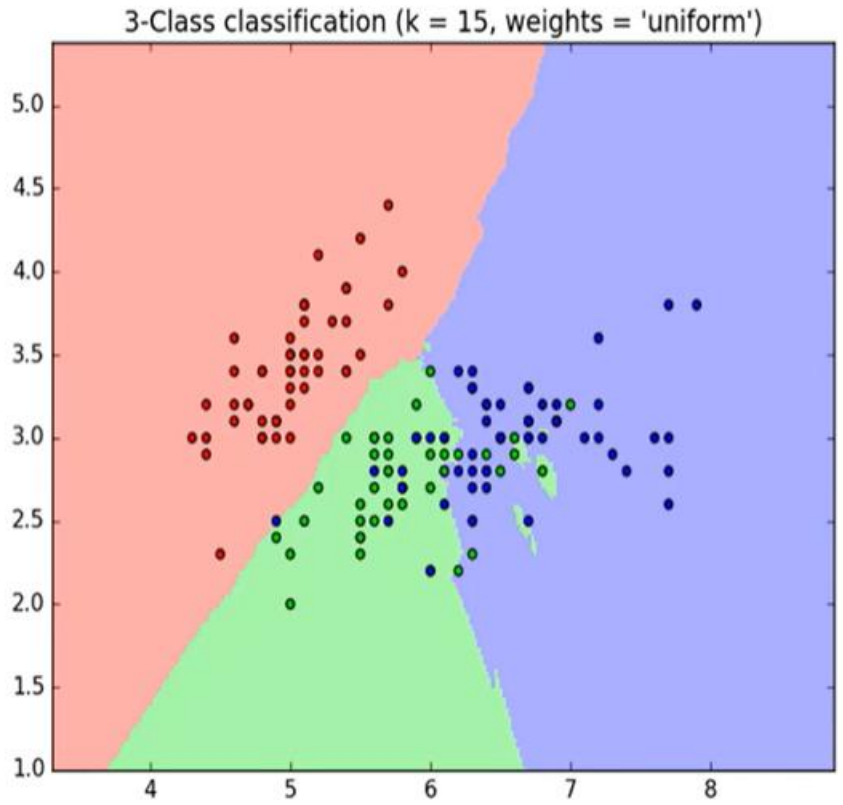
This is a sample example , Using **Scikit-learn packages with few lines of code** to build a **machine learning model**.


Machine learning categories
Supervised Machine Learning
Un Supervised Machine Learning
Semi Supervised Machine Learning

Supervised Learning



What is supervised learning?



We “teach the model,” then with that knowledge, it can predict unknown or future instances.

How we will Teach the Model ?

Teaching the model with labeled data

This example is taken from the cancer dataset.

ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000025	5	1	1	1	2	1	3	1	1	benign
1002945	5	4	4	5	7	10	3	2	1	benign
1015425	3	1	1	1	2	2	3	1	1	malignant
1016277	6	8	8	1	3	4	3	7	1	benign
1017023	4	1	1	3	2	1	3	1	1	benign
1017122	8	10	10	8	7	10		7	1	malignant
1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

The names up here, which are called **Clump thickness**, **Uniformity of cell size**, **Uniformity of cell shape**, **Marginal adhesion**, and so on, are called **Attributes or features.**

When dealing with **machine learning**, the most commonly used **data** is **numeric** and **categorical**

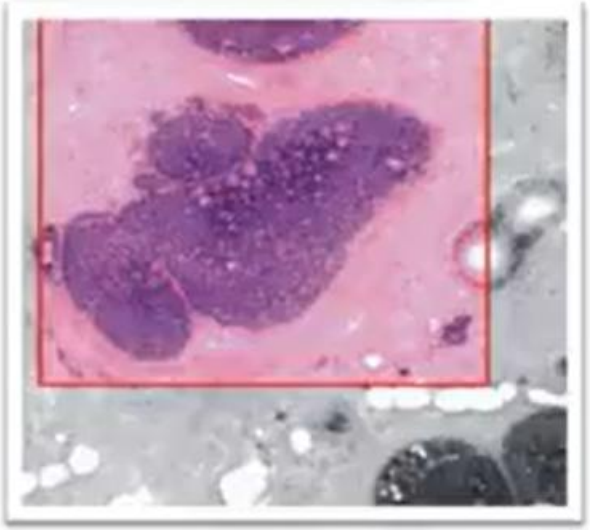
Teaching the model with labeled data

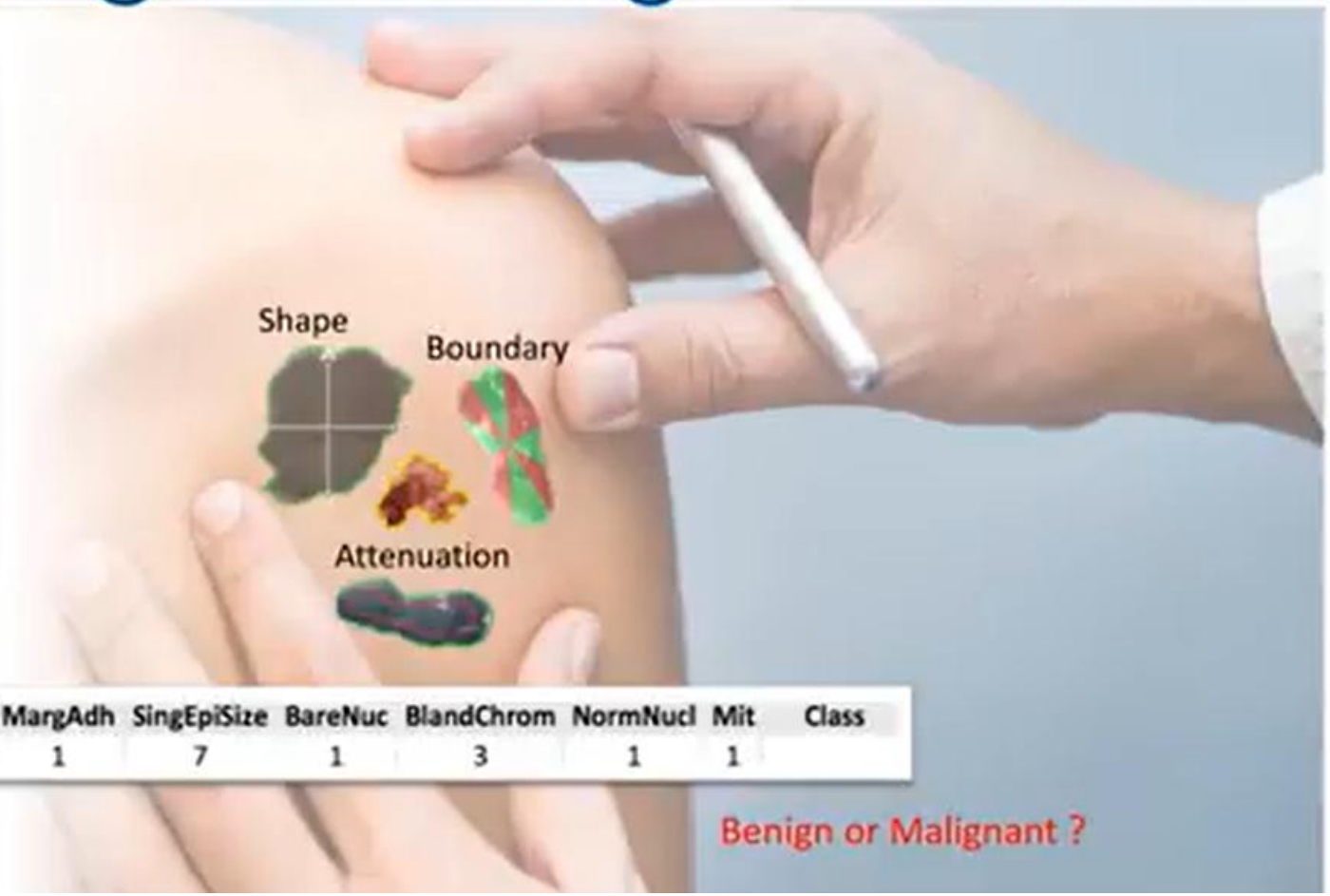
ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
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1018099	1	1	1	1	2	10	3	1	1	benign
1018561	2	1	2	H	2	1	3	1	1	benign
1033078	2	1	1	1	2	1	1	1	5	benign
1033078	4	2	1	1	2	1	2	1	1	benign

When dealing with classification problems

The data used to make predictions **(features)** can be either **numeric or categorical**, but the output (the class) is **categorical.**

Is this a benign or malignant cell?

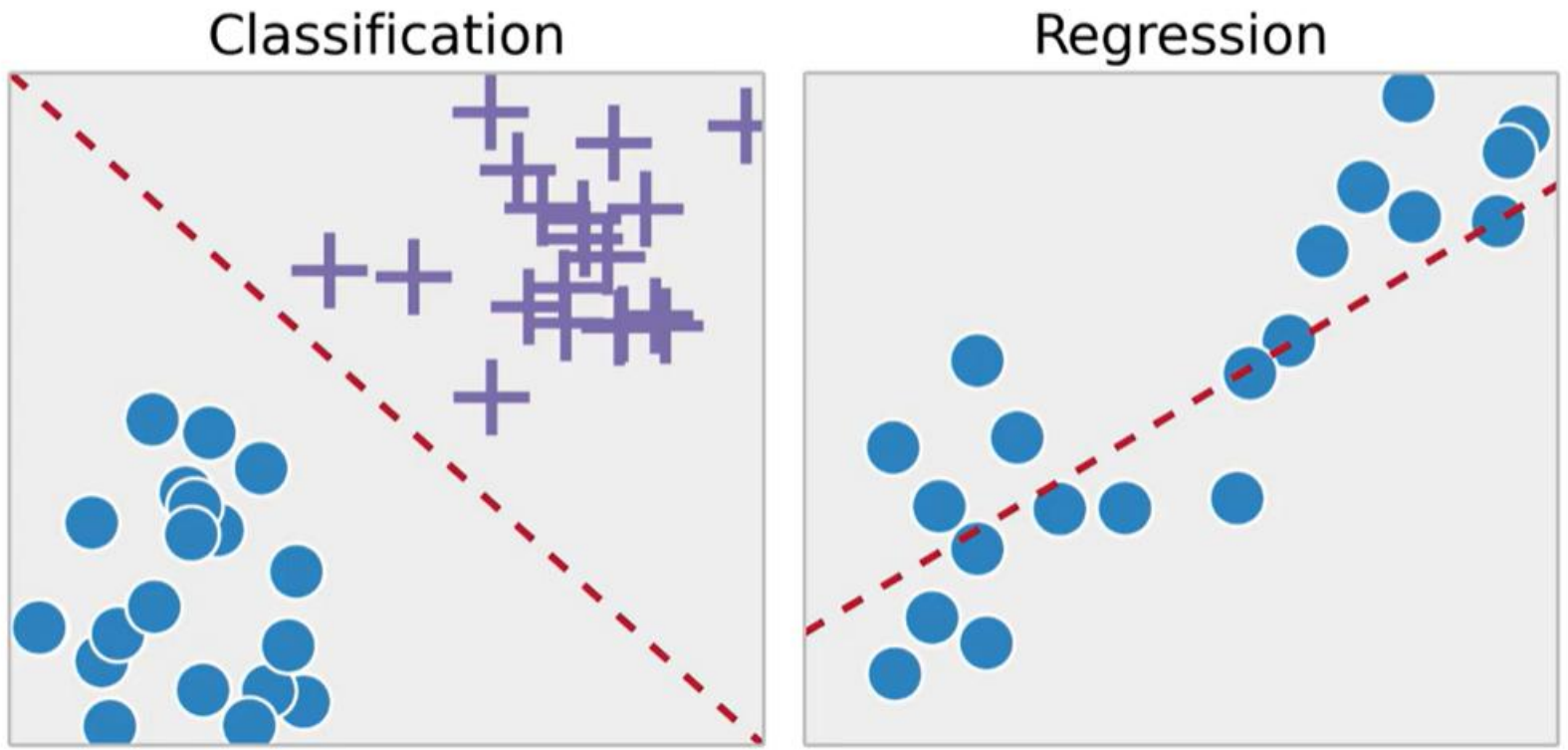




ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	BlandChrom	NormNucl	Mit	Class
1000015	6	1	1	1	7	1	3	1	1	

Benign or Malignant ?

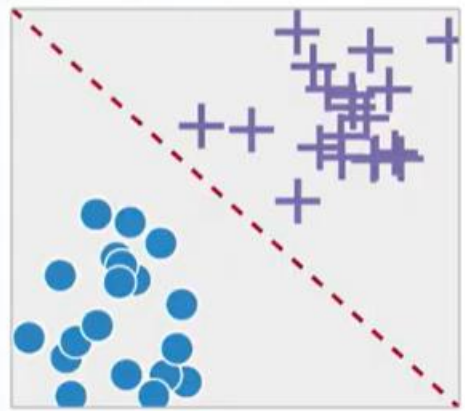
Types of supervised learning



What is classification?

Classification is the process of predicting discrete class labels or categories.

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1000025	5	1	1	1	2	1	3	1	1	benign
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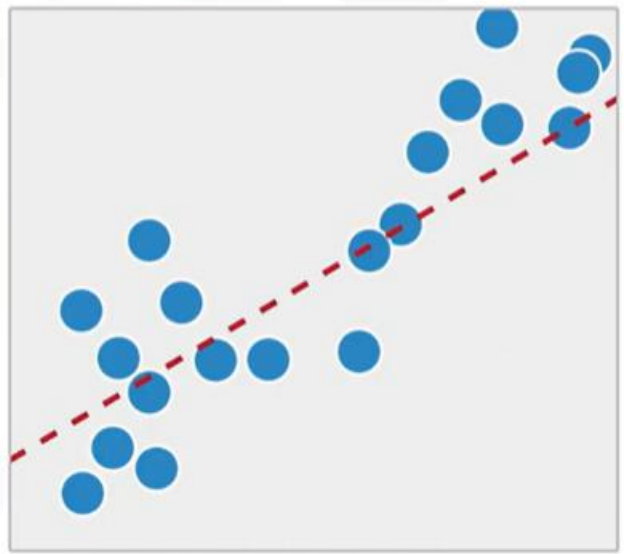


What is regression?

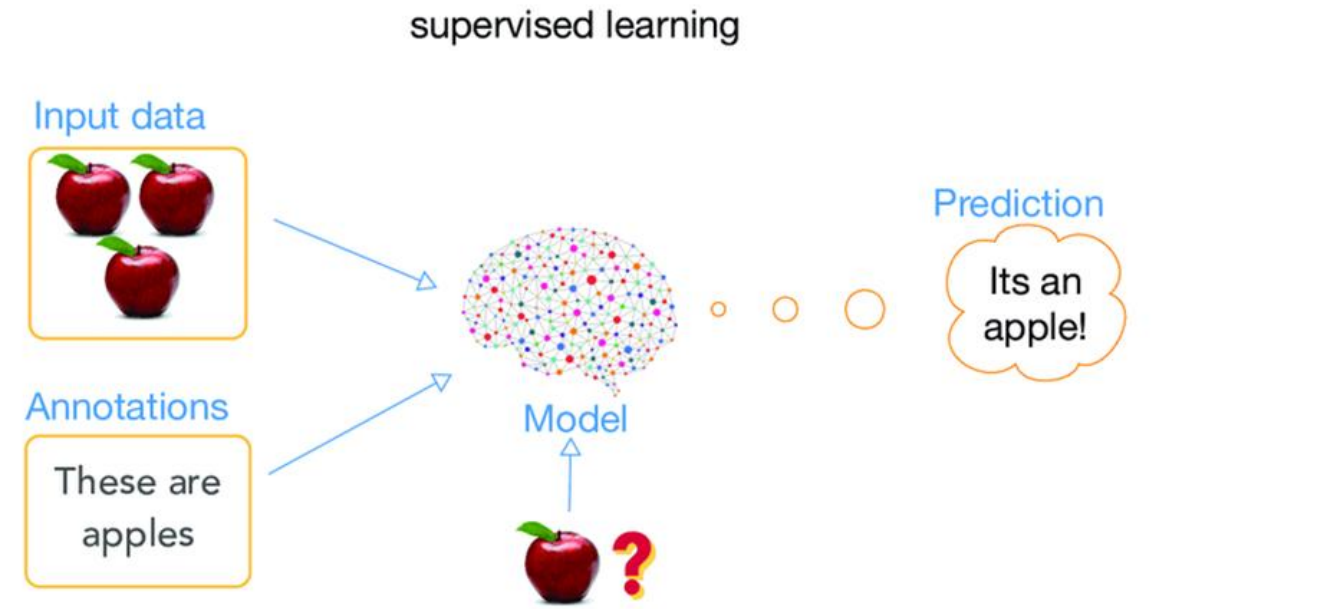
Regression is the process of predicting continuous values.

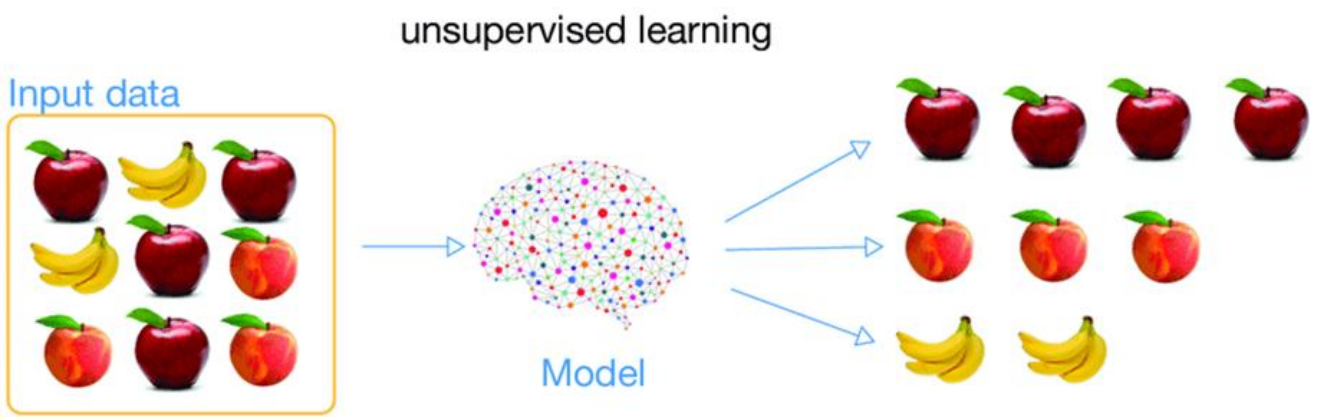
	ENGINESIZE	CYLINDERS	FUELCONSUMPTION_COMB	CO2EMISSIONS
0	2.0	4	8.5	196
1	2.4	4	9.6	221
2	1.5	4	5.9	136
3	3.5	6	11.1	255
4	3.5	6	10.6	244
5	3.5	6	10.0	230
6	3.5	6	10.1	232
7	3.7	6	11.1	255
8	3.7	6	11.6	267
9	2.4	4	9.2	?

Continuous Values



Supervised vs Unsupervised





What is unsupervised learning?

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
1	41	2	6	19	0.124	1.073	NBA001	6.3
2	47	1	26	100	4.582	8.218	NBA021	12.8
3	33	2	10	57	6.111	5.802	NBA013	20.9
4	29	2	4	19	0.681	0.516	NBA009	6.3
5	47	1	31	253	9.308	8.908	NBA008	7.2
6	40	1	23	81	0.998	7.831	NBA016	10.9
7	38	2	4	56	0.442	0.454	NBA013	1.6
8	42	3	0	64	0.279	3.945	NBA009	6.6
9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6



ALL OF THIS DATA
IS UNLABELED

The model works on its own to discover information.

We do not **supervise the model**, but we let the model **work on its own to discover insights** that may **not be visible to the human eye**.

Reducing the number of **features or variables** in a dataset while **retaining** its essential information.

What is unsupervised learning?

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio
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9	26	1	5	18	0.575	2.215	NBA006	15.5
10	47	3	23	115	0.653	3.947	NBA011	4
11	44	3	8	88	0.285	5.083	NBA010	6.1
12	34	2	9	40	0.374	0.266	NBA003	1.6

Unsupervised learning techniques:

- Dimension reduction
- Density estimation
- Market basket analysis
- Clustering

ALL OF THIS DATA IS UNLABELED

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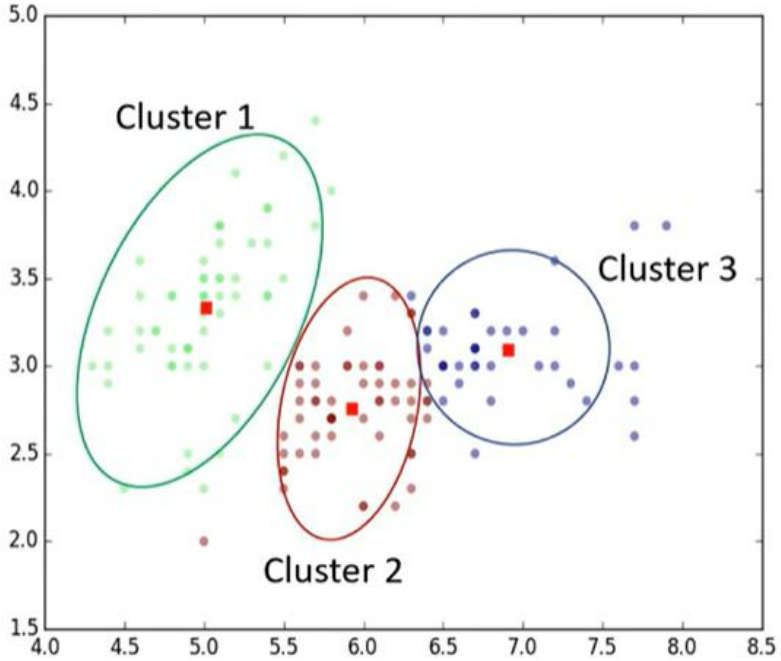
Figuring out how data is **spread out or distributed**, allowing us to identify **patterns and anomalies**.

Example : A **bank's** desire to **segment** its customers based on **certain characteristics**

What is clustering?

Clustering is grouping of data points or objects that are somehow similar by:

- Discovering structure
- Summarization
- Anomaly detection



- Uncovering **patterns and structures** within data by grouping similar items together.
- It helps **summarize** large datasets by **organizing them into meaningful clusters**.
- Additionally, clustering can **identify outliers** or anomalies by highlighting data points that do not fit well into any cluster.

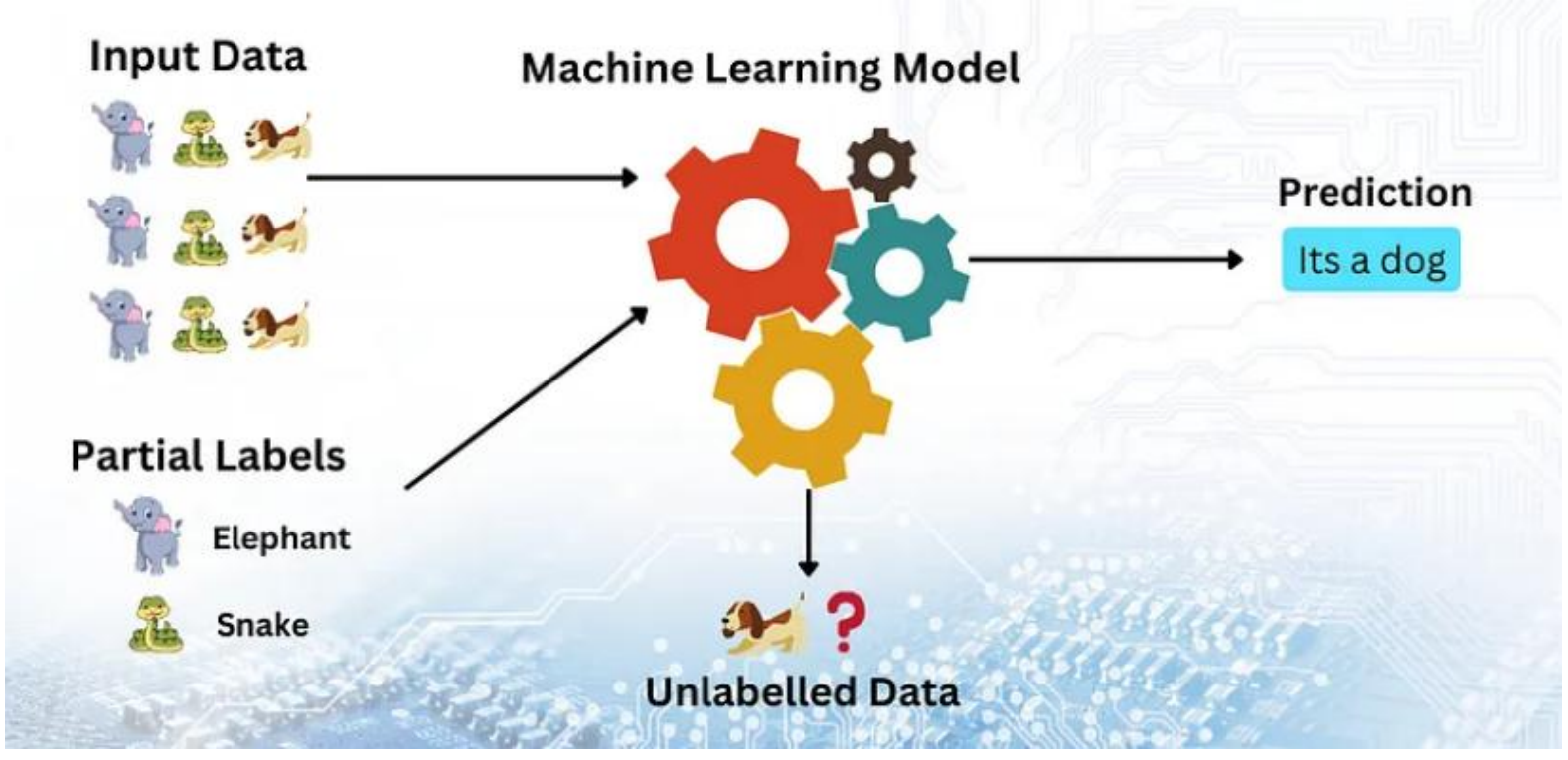
What is Semi Supervised Learning

- **Semi-supervised learning** is a type of machine learning that falls between **supervised and unsupervised learning**.
- In this algorithm, **training data** is a combination of **both labeled and unlabeled data**
- It uses a **large amount of unlabeled data** along with a **small amount of labeled data** to improve learning accuracy.

.....

- The main goal is to make the **most out of the available data** when **labeling** is **expensive or time-consuming**.
- Commonly used in scenarios where **unlabeled data is abundant**, such as in **image classification or text classification tasks.**

Ex: Semi-Supervised Support Vector Machines



What is Reinforcement Learning?

- **Reinforcement Learning** is a **feedback-based Machine learning technique** in which an agent **learns to behave in an environment** by performing the actions and seeing the results of actions.
- For each **good action**, the agent gets **positive feedback**, and for each **bad action**, the agent gets **negative feedback** or penalty.

.....

- Since there is **no labeled data**, so the **agent** is bound to **learn by its experience only.**
- The primary goal of an **agent** in reinforcement learning is to improve the performance by getting the maximum positive rewards.
- Reinforcement Learning (RL) plays a significant role in the development of **autonomous vehicles**, also known as **self-driving cars.**

